

Group I, Row V (the loop row): check, UV, and the p^+ integration

Your Group_V.tex and rc_term.pdf, checked line by line — and one question I need answered before going further

Notation of your rc_term.pdf: $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = \frac{p^+}{k^+}$, $q = \bar{\xi}k$, $D = (r - s)^2 + M^2$, $M^2 = \frac{\xi}{\bar{\xi}}s^2$, $D_0 = \frac{s^2}{\xi}$, $\mathcal{W} = \frac{P \cdot s}{P^2 s^2}$, $L = \log \frac{k^+}{\Lambda}$, $C_{UV}(\xi) = \xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{\mathcal{P}_{gg}(\xi)}{2N_c}$, $\mathcal{N} = \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+}$

In one box.

1. **Group_V.tex is correct** in every equation (six tensor blocks, prefactor, $\delta^{(2)}$, the w integral, the six p^+ weights, the colour rewriting, the identity, the split). Only placement slips.
2. **Your UV result (1.22) in rc_term.pdf is correct**, prefactor $\mathcal{N}$ included. Two of the term-by-term rewritings, **(1.8) and (1.9), are wrong**, and (1.14) $\rightarrow$ (1.15) drops a finite term; none of this touches the pole.
3. **Part (a) has no UV divergence**.
4. **" $I(\xi)$ is a finite term" is only half true**. It has no $\frac{1}{\epsilon}$, but at $\xi \rightarrow 1$ it carries $-\frac{\pi\mathcal{W}}{\xi} \log \frac{1}{\xi}$, which **halves** the $\log^2 \frac{k^+}{\Lambda}$ of (1.22), and it means a single plus prescription is **not** enough at that end.
5. At $\xi \rightarrow 0$ the plus prescription is legitimate at fixed x, y — but the $\delta(\xi)$ term it produces is **logarithmically divergent in the remaining $\int_x$** (a soft gluon far away). How to treat that is a choice about how the whole calculation is organised, so **I have stopped there**: see §7.

Corrections to my own recent answers. This is the same row I checked in an earlier session, and those (numerically verified) results agree with your note. Three things I told you in the last few answers were wrong and are corrected here: (i) my part-(b) UV formula was missing the $\frac{1}{k^+}$ — your $\mathcal{N}$ is right; (ii) I said the part-(b) remainders need only a one-sided plus prescription — at $\xi \rightarrow 1$ they need two distributions (§5); (iii) for part (a) I claimed a $\frac{\log}{\xi}$ at $\xi \rightarrow 0$ as well — in your order of integration (r first, at fixed x, y) there is none; that log moves into $\int_x$ (§5, §7). My $\frac{1}{\epsilon}$ also carried the opposite sign convention to yours; the physics is identical, and I use yours from here on.

1. Check of Group_V.tex

All verified numerically against the quoted $\mathfrak{A}^{(1)}$, $\mathfrak{B}_2$, $\mathfrak{C}_1$:

- The six tensor blocks equal $\delta_{im} \text{tr} B - \frac{k^+}{p^+} B_{mi} - \frac{k^+}{k^+ - p^+} B_{im}$ to 3×10^{-14} in $d_\perp = 2, 3, 4, 5$; the $d_\perp$ is δ^{jj} from the trace.
- Prefactor $\frac{g^4}{8\pi^4} \frac{p^+(k^+ - p^+)}{k^{+3}}$ exact. The vertex product alone gives a *minus*; the $+$ you write needs the first bracket to supply the second minus, which rests on $\mathfrak{A}$ — still not defined in the file.
- $\delta^{(2)}$ sets $w = z - \xi(z - x)$; the two phases and all six p^+ weights after the w integral are exact.
- $f^{ebf} = -f^{fbe}$ and $-f^{abc} f^{cbd} U^{df} = +N_c U^{af}$ exact; the identity holds with $S^{ad}(z) = U^{ad}(z)$ (both lines at the *same* point); the split into (a)+(b) is exact to 10^{-15} .

Placement slips (all equations from line 60 on): the phases $e^{-ik \cdot (w' - z)} e^{-ik \cdot (z - x) p^+ / k^+}$ and the colour factors sit outside the $\int dp^+$ and $\int_{x,y,z,w'}$ they depend on, and one $\left\{ \right.$ is closed twice. I would also write part (a)'s colour as $-f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)]$ (identical, and its UV finiteness is then visible).

2. Check of rc_term.pdf

eq.	status	detail
(1.2)	correct	$\frac{1}{k^+ \xi D}, M^2 = \frac{\xi}{s} s^2$.
(1.3)	typo	the middle expression lacks the $\frac{p^+}{k^+}$; the final form with $q = \bar{\xi} k$ is right.
(1.4)–(1.7), (1.10)	correct	agree with the exact contraction to 10^{-12} .
(1.8)	wrong	$[1 - \frac{r \cdot s}{s^2}]$ must be $[1 - \frac{r \cdot s}{2s^2}]$ (off by 25 as written, 10^{-13} fixed).
(1.9)	wrong	$\frac{s^2 - s \cdot r}{2(s-r)^2}$ must be $\frac{s^2 - s \cdot r}{2r^2(s-r)^2}$ (off by 11 as written, 10^{-12} fixed).
(1.12), (1.13)	inherit	the two errors above; they do <i>not</i> affect the UV part (both terms are subleading at $r \rightarrow 0$).
(1.14) $\rightarrow$ (1.15)	drops a finite term	replacing $\frac{(P \cdot r)(s \cdot r)}{r^4} \rightarrow \frac{P \cdot s}{d_\perp r^2}$ is exact for the pole and the log, but misses the traceless part (§3).
(1.17)	typo	a stray $\int d^2 r$ next to $\int d^2 l$ (the r integral is already done).
(1.18)	typo	$\Gamma[d-2] = \Gamma[-2\epsilon]$, not $\Gamma[1-2\epsilon]$; your final $-\frac{2}{\epsilon} \frac{\Gamma[1+\epsilon]\Gamma[1-\epsilon]^2}{\Gamma[1-2\epsilon]}$ is right.
(1.19)–(1.22)	correct	see §4.

3. Part (b): the UV terms

The UV region is $r \rightarrow 0$ ($z \rightarrow x$). The exact singular part of $T(r)$ is

$$T_{\text{UV}}(r) = d_\perp \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2 r^2},$$

and $\frac{1}{\xi D} \rightarrow \frac{1}{\xi D_0} = \frac{1}{s^2}$. Block by block: $T_1 \rightarrow \xi \bar{\xi}$, $T_2 \rightarrow \frac{\xi}{\bar{\xi}}$, $T_4 \rightarrow \frac{\bar{\xi}}{\xi}$ — one term of C_{UV} each — while the T_3, T_5, T_6 pieces cancel *pointwise* because their coefficient is $2 - 2\xi - 2\bar{\xi} = 0$. Checked: $\rho^2 \times \langle T/(\bar{\xi} D) \rangle \rightarrow C_{\text{UV}} \mathcal{W}$ to 8 digits at five values of ξ , and the subtracted integrand falls as ρ^2 .

The two masters (in $d = 2 - 2\epsilon$, your $\overline{\text{MS}}$ convention)

$$\int d^d r \frac{e^{iq \cdot r}}{r^2} = \pi \left[\frac{1}{\epsilon} + \log \frac{\mu_{\overline{\text{MS}}}^2}{q^2} \right], \quad \int d^d r e^{iq \cdot r} \frac{r^i r^m}{r^4} = \frac{\pi}{2} \left[\frac{1}{\epsilon} + \log \frac{\mu_{\overline{\text{MS}}}^2}{q^2} \right] \delta^{im} - \pi \hat{q}^i \hat{q}^m.$$

The first is your (1.19)–(1.20). The second is where (1.15) loses something: the $-\pi \hat{q}^i \hat{q}^m$ is finite, and contracting with $d_\perp \xi \bar{\xi} P^i s^m$ (the $\epsilon \cdot \frac{1}{\epsilon}$ from $d_\perp = 2 - 2\epsilon$ included) leaves

$$\int d^d r e^{iq \cdot r} \frac{T_{\text{UV}}(r)}{s^2} = \pi \mathcal{W} C_{\text{UV}}(\xi) \left[\frac{1}{\epsilon} + \log \frac{\mu_{\overline{\text{MS}}}^2}{\xi^2 k^2} \right] + \pi \xi \bar{\xi} \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2},$$

with $\hat{q} = \hat{k}$. Cutoff cross-check: the $\hat{q} \hat{q}$ coefficient comes out $-3.14159265359 = -\pi$.

4. Part (b): the answer

$$\frac{d^3 N_b^{\text{NLO}}}{d^3 k} = \mathcal{N} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \int_\lambda^{1-\lambda} d\xi \left\{ \mathcal{W} C_{\text{UV}}(\xi) \left[\frac{1}{\epsilon} + \log \frac{\mu_{\overline{\text{MS}}}^2}{\xi^2 k^2} \right] + \xi \bar{\xi} \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2} + \frac{1}{\pi} \mathcal{I}(\xi) \right\} \mathcal{U}_b$$

$$\mathcal{I}(\xi) = \int d^2 r e^{i\bar{\xi} k \cdot r} \left[\frac{T(r)}{\bar{\xi} D} - \frac{T_{\text{UV}}(r)}{s^2} \right]_{d_\perp=2}, \quad \mathcal{U}_b = [U^{ab'}(x') - U^{ab'}(w')] N_c [U^{af}(y) - U^{af}(x)] \rho^{b'}(x') \rho^f(y)$$

with $T(r)$ the six blocks, (1.8) and (1.9) corrected. $\mathcal{I}$ has no pole and needs no regulator.

The first two terms integrate in closed form ($\lambda = \frac{\Lambda}{k^+}$; each confirmed by quadrature):

$$\int_\lambda^{1-\lambda} C_{\text{UV}} d\xi = 2L - \frac{11}{6}, \quad \int_\lambda^{1-\lambda} C_{\text{UV}} \log \frac{1}{\xi^2} d\xi = L^2 + \frac{\pi^2}{3} - \frac{67}{18}, \quad \int_0^1 \xi \bar{\xi} d\xi = \frac{1}{6},$$

which gives exactly your (1.22) for the first term, plus a new finite $\frac{1}{6} \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2}$ from the second. The $-\frac{11}{6}$ is $\frac{\beta_0}{2N_c}$ at $n_f = 0$.

5. Part (b): can the plus prescription do the remaining ξ integral?

Everything hinges on how $\mathcal{I}(\xi)$ behaves at the two endpoints, at fixed x, y (your order: r first). From the 2D oscillatory quadrature of the earlier session:

endpoint	behaviour of $\mathcal{I}$	numerical evidence	plus prescription
$\xi \rightarrow 0$	$\frac{b_0}{\xi}$, simple pole	$\xi \text{ Re } \mathcal{I}$: 2.6020, 2.6143, 2.6154 at $\xi = 10^{-2}, -3, -4$ (slope $\rightarrow 0$)	legitimate: $g(0)$ finite
$\xi \rightarrow 1$	$\frac{-\pi \mathcal{W} \log \frac{1}{\xi} + b_1}{\xi}$	$\xi \text{ Re } \mathcal{I}$ from 11.56 to 29.24 as $\xi: 3 \cdot 10^{-2} \rightarrow 10^{-4}$, slope 3.061 = $-\pi \mathcal{W}$ (predicted 3.061039)	not with one distribution: $g(1)$ diverges

Why there is a log at $\xi \rightarrow 1$ — three scales

As $\bar{\xi} \rightarrow 0$ the r integral has three scales: $|s|$, the mass $M = |s| \sqrt{\xi/\bar{\xi}} \rightarrow \infty$, and the oscillation length $\frac{1}{q} = \frac{1}{\xi k} \gg M$. The true integrand has a $\frac{1}{r^2}$ tail in $|s| < |r| < M$ (from T_2 , whose weight $\frac{\xi}{\bar{\xi}}$ is large) and another in $M < |r| < \frac{1}{q}$ (from the T_3, T_4, T_6 pieces that grow like $|r|$). Each gives $\frac{\pi \mathcal{W}}{2\xi} \log \frac{1}{\xi}$; together $+\frac{\pi \mathcal{W}}{\xi} \log \frac{1}{\xi}$. The UV counterterm $\frac{T_{UV}}{s^2}$, however, keeps its $\frac{1}{r^2}$ all the way from 0 to $\frac{1}{q}$ and so carries $\frac{2\pi \mathcal{W}}{\xi} \log \frac{1}{\xi}$ — that is the $\log \frac{1}{\xi^2}$ in (1.21). The remainder $\mathcal{I} = \text{true} - \text{counterterm}$ therefore has $-\frac{\pi \mathcal{W}}{\xi} \log \frac{1}{\xi}$.

The correct prescription at $\xi \rightarrow 1$

$$\frac{1}{\bar{\xi}} = \left[\frac{1}{\bar{\xi}} \right]_+ + \delta(\bar{\xi}) L, \quad \frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} = \left[\frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} \right]_+ + \frac{1}{2} \delta(\bar{\xi}) L^2.$$

Applied to $\frac{1}{\pi} \mathcal{I}$ it contributes $-\frac{1}{2} \mathcal{W} L^2 + \frac{b_1}{\pi} L$ to the braces, so the double logarithm of part (b) is

$$\left(1 - \frac{1}{2}\right) \mathcal{W} L^2 = \frac{1}{2} \mathcal{W} \log^2 \frac{k^+}{\Lambda}, \quad \text{not } \mathcal{W} \log^2 \frac{k^+}{\Lambda} \text{ as (1.22) alone suggests.}$$

At $\xi \rightarrow 0$

$\left[\frac{1}{\xi}\right]_+$ alone is legitimate at fixed x, y and gives $\delta(\xi) \frac{b_0(x, y)}{\pi} L$. But see §7 before using it.

6. Part (a): no UV; the same endpoint structure

No UV divergence. $\mathcal{C}_a = -f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)] = O(|r|)$, and that linear piece is odd in $\hat{r}$ while the UV residue is even, so two powers are gained: with a smooth $U(z)$ field, $\rho^2 \langle \cdot \rangle$ is $2.1 \times 10^{-5}, 10^{-7}, 10^{-9}, 10^{-11}$ at $|r| = 10^{-2} \dots 10^{-5}$ against the colour-stripped constant 0.0134814. Keep $d_\perp = 2$; nothing to regularise.

The ξ integral has the same endpoint structure as part (b), with no counterterm to subtract:

- $\xi \rightarrow 0$: simple pole at fixed x, y (the only $\frac{1}{\xi}$ is T_4 , which vanishes at $r = s$, and the large- $|r|$ tail oscillates as $e^{ik \cdot r}$). $\left[\frac{1}{\xi}\right]_+$ legitimate — again, see §7.
- $\xi \rightarrow 1$: $\frac{+\pi \mathcal{W} \mathcal{C}_a^\infty \log \frac{1}{\xi} + b_1^{(a)}}{\xi}$, with $\mathcal{C}_a^\infty = -f^{adc} f^{fbe} U^{db}(x) + N_c U^{af}(x)$ the colour at large $|r|$. Two distributions, giving $+\frac{1}{2} \mathcal{W} \mathcal{C}_a^\infty L^2$. This assumes $U(z) \rightarrow 1$ outside the target, the usual finite-target convention; $\|\mathcal{C}_a^\infty\| = 1.30$ in the test configuration, so it does not vanish.

7. Where I stopped, and why

The $\delta(\xi)$ term from $\xi \rightarrow 0$ is divergent in the remaining $\int_x$. At $\xi = 0$ the total phase $\exp[-ik \cdot (w' - \bar{\xi}z - \xi x)]$ loses all dependence on x , and the residue of the $\frac{1}{\xi}$ pole has a $\frac{1}{x^2}$ tail at large $|x|$ (fixed y, z). Checked: $|x|^2 \langle \text{residue} \rangle \rightarrow -0.0457352$ at $|x| = 10^{1 \dots 4}$, matching $\frac{P \cdot (y-z)}{2P^2(y-z)^2} = -0.0457352$. So $\int d^2x$ of the $\delta(\xi)$ coefficient diverges logarithmically. Physically: the gluon with $p^+ \rightarrow 0$ sitting far from the others — the mirror image, under $(x \leftrightarrow z, \xi \leftrightarrow \bar{\xi})$, of the $\xi \rightarrow 1$ log in §5.

Its colour factor at large $|x|$ is $N_c[U(y) - 1]$ in part (b) and $-f^{adc}f^{fde}[U^{ce}(z) - \delta^{ce}]$ in part (a). Neither is the LO colour structure, so this piece cannot be absorbed inside this row. There are two consistent ways forward, and they lead to *different-looking* final answers for this row:

	what it means	what the answer looks like
(A)	Regulate the soft- x region <i>within this row</i> : keep ξ finite through the $\int_x$, so its own oscillation $e^{i\xi k \cdot x}$ cuts it off at $ x \sim \frac{1}{\xi k}$.	A $\frac{\log(1/\xi)}{\xi}$ at $\xi \rightarrow 0$ as well; two distributions at <i>each</i> end; an extra $\frac{1}{2}L^2$ with colour $N_c[U(y) - 1]$ (b) and $-ff[U(z) - 1]$ (a).
(B)	Leave it for the <i>cross-row</i> soft subtraction: combine the $U(x) \rightarrow 1$ pieces of this row with those of the other Group I rows (e.g. Row IV) before any ξ integral, where they should cancel.	This row keeps only the $\xi \rightarrow 1$ double log; the $\delta(\xi)$ coefficient is carried unintegrated in x until the rows are combined.

Both are legitimate; which one fits depends on how you and your advisor are organising the soft/rapidity divergences across the whole NLO calculation. I did not want to pick one and hand you a “final answer” built on the wrong choice.

Checks: `src/row28_full.py` (the `rc_term` rewritings (1.4)–(1.10), the $r \rightarrow 0$ residue, the two Fourier masters, the ξ moments), `src/row28_rem.py` (the endpoint scans of $\mathcal{I}$), `src/rowB2_check.py`, `src/rowB2_split.py`, `src/rowB2_identity.py` (the `.tex` equations), `src/rowB2_2a_full.py` (part (a) UV), `src/rowV_softx.py` (the soft- x tail).